COS 201

COMPUTER PROGRAMMING I

Assignments and Practical Exercises — Weeks 1–3

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Matriculation Number	25/18467
Course	COS 201 — Computer Programming I
Programme	Computer Science
GitHub Repository	https://github.com/Amodu-png/COS-201

Prepared from the uploaded COS 201 Weeks 1–3 lecture note. Java solutions include sample outputs or execution notes where useful.

This document contains the Week 1 written assignments and the Week 2?3 Java practical exercises, with sample outputs and text-based flowcharts.

Week 1 — Activity A: Identify the IPO components

1. Rectangle area (length 12, width 5)

Input: length = 12, width = 5.

Process: area = length × width = 12 × 5 = 60.

Output: area = 60 square units.

2. Average of four examination scores

Input: score1, score2, score3, score4.

Process: total = score1 + score2 + score3 + score4; average = total ÷ 4.

Output: average examination score.

3. Classify a number

Input: number n.

Process: if n > 0 classify positive; else if n < 0 classify negative; otherwise classify zero.

Output: “Positive”, “Negative” or “Zero”.

4. Cost of six items at ■250 each

Input: quantity = 6, unit price = ■250.

Process: total = 6 × 250 = ■1,500.

Output: total price = ■1,500.

5. Failed login attempts in a sample log

Input: sample security log entries / login-result fields.

Process: inspect each entry and increment a counter when the result is failed; stop at end of log.

Output: count of failed login attempts.

Week 1 — Activity B: Algorithms

1. Add two numbers

```
START
INPUT first, second
sum ← first + second
OUTPUT sum
END
```

2. Area of a circle

```
START
INPUT radius
area ← π × radius × radius
OUTPUT area
END
```

3. Pass/fail using 50 as pass mark

```
START
INPUT score
IF score >= 50 THEN
    OUTPUT "Pass"
ELSE
    OUTPUT "Fail"
END IF
END
```

4. Largest of three numbers

```
START
INPUT a, b, c
largest ← a
IF b > largest THEN largest ← b
IF c > largest THEN largest ← c
OUTPUT largest
END
```

5. Total cost of items

```
START
INPUT quantity, unitPrice
totalCost ← quantity × unitPrice
OUTPUT totalCost
END
```

Week 1 — Activity C: Pseudocode and flowcharts

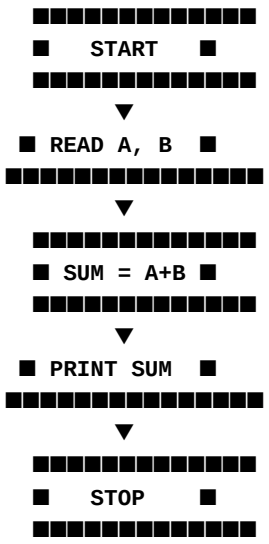
Selected problem 1: Add two numbers

Numbered algorithm: (1) Start. (2) Read two numbers A and B. (3) Compute SUM = A + B. (4) Display SUM. (5) Stop.

Pseudocode

```
BEGIN
    READ A, B
    SUM ← A + B
    PRINT SUM
END
```

Flowchart



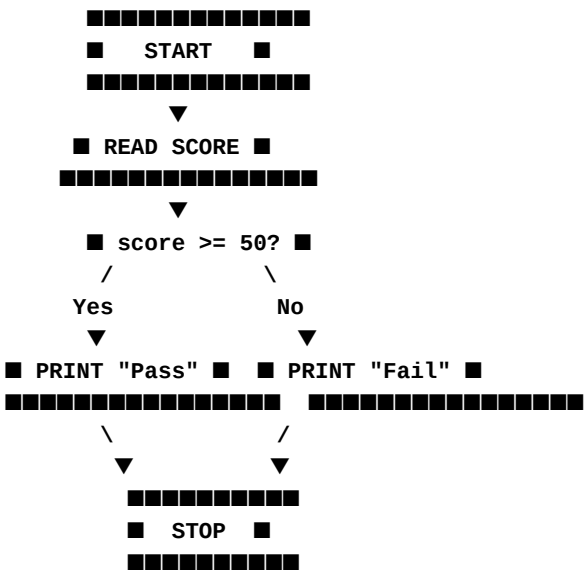
Selected problem 2: Determine pass or fail

Numbered algorithm: (1) Start. (2) Read score. (3) If score is at least 50, display Pass. (4) Otherwise display Fail. (5) Stop.

Pseudocode

```
BEGIN
    READ score
    IF score >= 50 THEN
        PRINT "Pass"
    ELSE
        PRINT "Fail"
    END IF
END
```

Flowchart



Week 1 — Practical: FirstProgram.java

This program displays the course information, programme of study, three reasons programming matters to the discipline, and a learning goal. The programme name and goal are written as editable student-specific text.

```
public class FirstProgram {
    public static void main(String[] args) {
        System.out.println("Course: COS 201 - Computer Programming I");
        System.out.println("Programme: Computer Science");
        System.out.println("Reason 1: Programming helps solve computational problems.");
        System.out.println("Reason 2: Programming supports automation of tasks.");
        System.out.println("Reason 3: Programming enables software development.");
        System.out.println("My goal: I want to develop simple working applications.");
    }
}
```

Expected output

```
Course: COS 201 - Computer Programming I
Programme: Computer Science
Reason 1: Programming helps solve computational problems.
Reason 2: Programming supports automation of tasks.
Reason 3: Programming enables software development.
My goal: I want to develop simple working applications.
```

Week 2 — Practical: HelloWorld.java

A basic Java program that prints a greeting and demonstrates the class, main method and output statement.

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, COS 201!");
        System.out.println("My name is Amodu Qudus Ajadi             .");
        System.out.println("Matriculation Number: 25/18467 ");
    }
}
```

Expected output

```
Hello, COS 201!
My name is Amodu Qudus Ajadi             .
Matriculation Number: 25/18467
```

Week 2 — Student profile practical

```
import java.util.Scanner;

public class StudentProfile {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = input.nextLine();

        System.out.print("Enter your programme: ");
        String programme = input.nextLine();

        System.out.print("Enter your level: ");
        int level = input.nextInt();

        System.out.println("\n--- STUDENT PROFILE ---");
        System.out.println("Name: " + name);
        System.out.println("Programme: " + programme);
        System.out.println("Level: " + level);
        input.close();
    }
}
```

Week 3 — Practical 1: SimpleCalculator.java

Reads two numbers and displays their sum, difference, product and quotient. Division by zero is handled explicitly.

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the first number: ");
        double firstNumber = input.nextDouble();

        System.out.print("Enter the second number: ");
        double secondNumber = input.nextDouble();

        System.out.printf("Sum: %.2f%n", firstNumber + secondNumber);
        System.out.printf("Difference: %.2f%n", firstNumber - secondNumber);
        System.out.printf("Product: %.2f%n", firstNumber * secondNumber);

        if (secondNumber != 0) {
            System.out.printf("Quotient: %.2f%n",
                              firstNumber / secondNumber);
        } else {
            System.out.println("Division is undefined for a zero divisor.");
        }
        input.close();
    }
}
```

Sample run

```
Enter the first number: 20
Enter the second number: 4
Sum: 24.00
Difference: 16.00
Product: 80.00
Quotient: 5.00
```

Week 3 — Practical 2: AverageCalculator.java

```
import java.util.Scanner;

public class AverageCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter student name: ");
        String name = input.nextLine();

        System.out.print("Enter score for Course 1: ");
        double score1 = input.nextDouble();
        System.out.print("Enter score for Course 2: ");
        double score2 = input.nextDouble();
        System.out.print("Enter score for Course 3: ");
        double score3 = input.nextDouble();

        double total = score1 + score2 + score3;
        double average = total / 3.0;

        System.out.println("\n--- STUDENT RESULT ---");
        System.out.println("Student: " + name);
        System.out.printf("Total: %.2f%n", total);
        System.out.printf("Average: %.2f%n", average);
        input.close();
    }
}
```

Sample run

```
Enter student name: Amodu Qudus Ajadi
Enter score for Course 1: 70
```

```
Enter score for Course 2: 80
Enter score for Course 3: 90

--- STUDENT RESULT ---
Student: Amodu Qudus Ajadi
Total: 240.00
Average: 80.00
```

Week 3 — Practical 3: ItemCost.java

```
import java.util.Scanner;

public class ItemCost {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter item name: ");
        String itemName = input.nextLine();

        System.out.print("Enter quantity: ");
        int quantity = input.nextInt();

        System.out.print("Enter unit price: ");
        double unitPrice = input.nextDouble();

        double totalCost = quantity * unitPrice;

        System.out.println("\n--- PURCHASE SUMMARY ---");
        System.out.println("Item: " + itemName);
        System.out.println("Quantity: " + quantity);
        System.out.printf("Unit price: %.2f%n", unitPrice);
        System.out.printf("Total cost: %.2f%n", totalCost);
        input.close();
    }
}
```

Sample run

```
Enter item name: Exercise book
Enter quantity: 3
Enter unit price: 1500
--- PURCHASE SUMMARY ---
Item: Exercise book
Quantity: 3
Unit price: 1500.00
Total cost: 4500.00
```

Week 3 — Debugging exercise

Original expression: double result = firstNumber / secondNumber; When both operands are int, Java performs integer division before converting the result to double. Cast one operand to double to obtain a fractional result.

```
int firstNumber = 5;
int secondNumber = 2;
double result = (double) firstNumber / secondNumber;
System.out.println(result); // 2.5
```

Practical submission messages

Week 2 submission draft

```
COS 201 — Week 2 Practical Submission

Dear Lecturer,

I am submitting my Week 2 practical assignment for COS 201 — Computer Programming I.

Student Name: Amodu Qudus Ajadi
Matriculation Number: 25/18467
Programme: Computer Science
Assignment: Development Environment and First Java Program
GitHub Repository: [Insert your actual repository URL]

I have included my Java source code and README documentation in
the repository. I have compiled and executed the program to verify
the output.

Thank you.
```